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# Step 1: Import Required Libraries

from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.naive_bayes import GaussianNB

from sklearn.metrics import confusion_matrix, accuracy_score, classification_report

# Step 2: Load Dataset

data = load_iris()

# Features and Target

X = data.data

y = data.target

# Step 3: Split Dataset into Training and Testing Data

X_train, X_test, y_train, y_test = train_test_split(

    X,

    y,

    test_size=0.2,

    random_state=42

)

# Step 4: Create Naïve Bayes Model

model = GaussianNB()

# Step 5: Train the Model

model.fit(X_train, y_train)

# Step 6: Predict Test Data

y_pred = model.predict(X_test)

# Step 7: Display Actual and Predicted Values

print("===== Actual Output =====")

print(y_test)

print("\n===== Predicted Output =====")

print(y_pred)

# Step 8: Display Confusion Matrix

cm = confusion_matrix(y_test, y_pred)
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print("\n===== Confusion Matrix =====")

print(cm)

# Step 9: Display Accuracy

accuracy = accuracy_score(y_test, y_pred)

print("\n===== Accuracy =====")

print("Accuracy = {:.2f}%".format(accuracy * 100))

# Step 10: Display Classification Report

print("\n===== Classification Report =====")

print(classification_report(

    y_test,

    y_pred,

    target_names=["Rose", "Jasmine", "Sunflower"]

))

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...  ===== Actual Output =====
      [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]

      ===== Predicted Output =====
      [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]

      ===== Confusion Matrix =====
      [[10  0  0]
       [ 0  9  0]
       [ 0  0 11]]

      ===== Accuracy =====
      Accuracy = 100.00%

      ===== Classification Report =====

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	precision	recall	f1-score	support
Rose	1.00	1.00	1.00	10
Jasmine	1.00	1.00	1.00	9
Sunflower	1.00	1.00	1.00	11
accuracy			1.00	30
macro avg	1.00	1.00	1.00	30
weighted avg	1.00	1.00	1.00	30